Techniques of Differentiation

1. Find the differential coefficients of the following functions with respect to x (i.e. $\frac{dy}{dx}$).

$(i)\, y = \sin x \sin 2x \sin 3x,\ \ (ii)\, y = \cos ec^3 x,\ \ (iii)\, y = \cos 2x \cos 3x,\ \ (iv)\, y = \sin^{-1}(x^2),$

$(v)\, y = \tan(\sin^{-1} x),\quad (vi)\cot^{-1}\left(\frac{1+x}{1-x}\right),\quad (vii)\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right),\quad (viii)\sin^{-1}\left(\frac{2x}{1+x^2}\right),$

$(ix)\tan^{-1}\left(\frac{2x}{1-x^2}\right),\quad (x)\tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right).$

2. Find the differential coefficients of:

$(i)(\sin x)^{\ln x},\qquad (ii)\,(\sin x)^{\cos x} + (\cos x)^{\sin x}.$

3. Find the differential coefficients of $\sqrt{\left(\frac{1+x}{1-x}\right)}$.

4. Find $\frac{dy}{dx}$ in the following cases:

$(i)\, 3x^4 - x^2 y + 2y^3 = 0,\quad (ii)\, x^3 + y^3 + 4x^2 y - 25 = 0,\quad (iii)\, x^y = y^x.$

5. Find $\frac{dy}{dx}$ when

$(i) x = a\cos^3\theta,\ y = a\sin^3\theta,\quad (ii) x = \sin^2\theta, y = \tan\theta,\quad (iii) x = a\sec^2\theta, y = a\tan^2\theta.$

6. Differentiate the left-side functions with respect to the right-side ones:

$(i)\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ with respect to $\tan^{-1}\left(\frac{2x}{1-x^2}\right)$ $\quad (ii) x^{\sin^{-1}(x)}$ with respect to $\sin^{-1} x$.

7. Find the differential coefficients of:

$(i)\ln\sqrt{\frac{1+\sin x}{1-\sin x}},\quad (ii)\ln\sqrt{\frac{1-\cos x}{1+\cos x}},\quad (iii)\tan^{-1}\sqrt{\frac{1-\cos x}{1+\cos x}},\quad (iv)\tan^{-1}\left(\frac{\cos x - \sin x}{\cos x + \sin x}\right)$

$(v)\sin\left(2\tan^{-1}\sqrt{\frac{1-x}{1+x}}\right).$

Answers

1. (i) $\cos x \sin 2x \sin 3x + 2\sin x \cos 2x \sin 3x + 3\sin x \sin 2x \cos 3x$
   (ii) $-3cosec^3 x \cdot \cot x$
   (iii) $-2\sin 2x \cos 3x - 3\cos 2x \sin 3x$
   (iv) $\frac{2x}{\sqrt{1-x^4}}$
   (v) Hint: Let $x = \tan\theta$. Ans: $\frac{2}{1+x^2}$
   (vi) Hint: Let $x = \tan\theta$. Ans: $\frac{2}{1+x^2}$
   (vii) Hint: Let $x = \sin\theta$. Ans: $\frac{1}{\sqrt{1-x^2}}$

2. (i) $(\sin x)^{\ln x}\left[\left(\frac{\ln(\sin x)}{x}\right) + \ln x \cot x\right]$

3. Hint: Use the quotient rule. Ans: $\frac{1}{(1-x)\sqrt{1-x^2}}$

4. (ii) $\frac{dy}{dx} = \frac{-3x^2-8xy}{3y^2+4x^2}$

5. (i) $-\tan\theta$

   (iii) 1

6. (i) 1

7. (ii) $\operatorname{cosec} x$

   (iii) Hint: $1 + \cos 2\theta = 2\cos^2\theta$, $1 - \cos 2\theta = 2\sin^2\theta$. Ans: $\frac{1}{2}$

   (iv)Hint: Divide by $\cos x$. Ans: -1

   (v) Hint: Let $x = \cos\theta$. Ans: $-\frac{x}{\sqrt{1-x^2}}$